TOR AMPLITUDE MODULO A NILPOTENT IDEAL AGREES WITH TOR AMPLITUDE OVER THE RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Tor amplitude of a complex and its behaviour under base change along a ring map. the author cites Lemma lemma-pull-tor-amplitude (tag 066L) at proof line 17315. Gives the easy direction of the equivalence.
- Derived tensor product of a complex with a module, computed by resolving the module. used at proof lines 17320-17335. Supplies the object whose cohomology is being bounded.
- Distinguished triangle attached to a short exact sequence of modules and the resulting long exact sequence in cohomology. used at proof lines 17330-17344. Is the tool that transfers the amplitude bound from the two outer terms to the middle one.

Source reference. Original lemma 0H75: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21445–21445. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 066L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7940–7940. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $R' \to R$ be a surjective ring map whose kernel is a nilpotent ideal. Let $K' \in D(R')$ and set $K = K' \otimes^{\mathbf{L}}_{R'} R$. Let $a, b \in \mathbf{Z}$. Then K has tor amplitude in $[a, b]$ if and only if K' does.*

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Proof. One direction follows from Lemma 066L. For the other, assume K has tor amplitude in $[a, b]$ and let M' be an R'-module. We have to show that $K' \otimes^{\mathbf{L}}_{R'} M'$ has nonzero cohomology only for degrees contained in the interval $[a, b]$.

Let $I = \mathrm{Ker}(R' \to R)$. Then $I^n = 0$ for some n. If $IM' = 0$, then we can view M' as an R-module and argue as follows

$$K' \otimes^{\mathbf{L}}_{R'} M' = K' \otimes^{\mathbf{L}}_{R'} (R \otimes^{\mathbf{L}}_{R} M') = (K' \otimes^{\mathbf{L}}_{R'} R) \otimes^{\mathbf{L}}_{R} M' = K \otimes^{\mathbf{L}}_{R} M'$$

which has nonvanishing cohomology only in the interval $[a, b]$ by assumption on K. If $I^{t+1}M' = 0$, then we consider the short exact sequence

$$0 \to IM' \to M' \to M'/IM' \to 0$$

By induction on t we have that both $K' \otimes^{\mathbf{L}}_{R'} IM'$ and $K' \otimes^{\mathbf{L}}_{R'} M'/IM'$ have nonzero cohomology only for degrees in the interval $[a, b]$. Then the distinguished triangle

$$K' \otimes^{\mathbf{L}}_{R'} IM' \to K' \otimes^{\mathbf{L}}_{R'} M' \to K' \otimes^{\mathbf{L}}_{R'} M'/IM' \to (K' \otimes^{\mathbf{L}}_{R'} IM')[1]$$

proves the same is true for $K' \otimes^{\mathbf{L}}_{R'} M'$ as desired. □